G. PALANI KUMAR REDDY

Embedded Systems and IoT Engineer

Bengaluru, Karnataka, India – +91 8247624835 – palanikumarreddy912@gmail.com

linkedin.com/in/palani-kumar-reddy-6443452a9 – github.com/Palani007-BTECHN

Summary

Embedded Systems and IoT Specialist with expertise in low-level microcontroller programming, microcode engineering, system board bring-up, and peripheral communication protocols. Hands-on experience developing firmware in Embedded C, C++, and Python across ESP32, PIC, ATmega328P, and Raspberry Pi platforms. Skilled in hardware validation, signal integrity analysis, and EDA software including Cadence Virtuoso, Proteus, and KiCad.

Work Experience

Conzura Soft Solutions Bengaluru, Karnataka

Embedded Engineer Intern *Jun 2024 – Aug 2024*

- Spearheaded system board bring-up and firmware validation for 8-bit and 32-bit hardware target platforms, increasing automated test coverage by 25%.
- Engineered 15+ modular unit, integration, and regression test suites in Embedded C to validate UART, SPI, and I2C peripheral protocols.
- Diagnosed low-level signal integrity faults and bus timing bugs using digital storage oscilloscopes and multi-channel logic analyzers.
- Authored technical validation logs, failure analysis documentation, and verification matrices across 4 major project cycles for engineering teams.

Projects

Smart Wildlife Conflict Prediction System – GitHub Link *Python, Streamlit, Folium, GPS/NavIC, LoRa, ML*

- Architected an end-to-end telemetry node integrating GPS/NavIC location modules with dynamic geofencing algorithms to predict wildlife entry zones.
- Designed a low power LoRa transmission node paired with a Streamlit GIS dashboard, achieving real-time location tracking accuracy within 5 meters.

RFID Automated Smart Billing System – GitHub Link *Arduino, RFID, Embedded C, LCD*

- Developed a multi-tag checkout system deploying 13.56 MHz RFID readers and ATmega microcontrollers, cutting checkout processing time by 40%.
- Programmed bare-metal driver software in Embedded C to parse card UID tokens, manage memory buffers, and output itemized billing metrics.

Agribot – Autonomous Fertilizer Sprayer & Grass Cutter – GitHub Link *Arduino, Embedded C, Motor Drivers, Robotics*

- Built an autonomous agricultural robot utilizing high torque DC motor drivers, PWM velocity control loops, and relay actuators.
- Conducted thermal profiling on power driver stages and optimized battery discharge cycles, expanding field operating runtime by 20%.

Autonomous Book Picking Library Robot *Arduino, RFID, Servo Controls, Ultrasonic Sensors*

- Engineered a mobile retrieval robot combining real-time ultrasonic obstacle detection with a 3 degree of freedom robotic arm, boosting inventory fetch speed by 30%.

8-bit Barrel Shifter Schematic & Simulation *Cadence Virtuoso, Digital Logic, CMOS*

- Modeled and simulated an 8-bit combinational Barrel Shifter in Cadence Virtuoso, optimizing signal propagation delay by 15% and reducing power dissipation.

IoT Environmental Monitoring System *Arduino, Wi-Fi, Environmental Sensors, Embedded C*

- Deployed a wireless telemetry node collecting 100+ daily data points to monitor ambient climate metrics and prevent temperature-sensitive cargo degradation.

Technical Skills

- **Programming Languages:** Embedded C, C, C++, Python
- **Hardware Platforms:** ESP32, PIC Microcontrollers, ATmega328P (Arduino Uno/Nano), Raspberry Pi 4
- **Communication Protocols:** UART, SPI, I2C, LoRa, RFID, Wi-Fi, Bluetooth, GSM
- **EDA & Tools:** Cadence Virtuoso, MPLAB X IDE, Proteus, KiCad, Arduino IDE, MATLAB, Git
- **Core Domains:** Firmware Engineering, Hardware-Software Integration, Circuit Simulation, PCB Design, IoT Architecture

EDUCATION

Jain (Deemed-to-be University) Bengaluru, Karnataka
Bachelor of Technology (B.Tech) in Electronics & Communication Engineering *Expected 2027*

- **Academic Standing:** CGPA 7.86 / 10.0
- **Core Coursework:** Embedded Systems, Microprocessors, VLSI Design, Digital Electronics, CMOS Analog Circuits

Sri Chaitanya Junior College Anantapur, Andhra Pradesh
Intermediate / Senior Secondary (MPC) – Score: 91.8% *Passed 2023*

Sri Chaitanya Techno School Anantapur, Andhra Pradesh
Secondary School Certificate (SSC) – Score: 100% *Passed 2021*

CERTIFICATIONS & LEADERSHIP

- **Technical Certifications:** IoT for Electronics (Coursera), Antennas & Wave Propagation (Coursera), Fundamentals of Electronics
- **Leadership Role:** Vice Treasurer, Evalanche Club (ECE Department) – Directed annual budget allocation and technical event logistics
- **Competitions:** Pitch Pulse Challenge, Forge & Flux Exhibition, Inceptrix Hackathon